cohesive (cos) ends. On attachment with tail to E.coli, phage injects its DNA into the cell. Inside E.coli the phage linear DNA cyclizes and gets ligated through cos ends to form a circular DNA .The phage DNA has two fates – lytic cycle and lysogenic cycle (Figure 12.27).

Only about 50% of phage λ DNA is necessary for its multiplication and other functions. Thus, as much as 50% (i.e. up to 20 kb) of the phage DNA can be replaced by a donor DNA for use in cloning experiments. However, several restriction sites are present on phage which is not by itself a suitable vector. The λ based phage vectors are modifications of the natural phage with much reduced number of restriction sites.

The main advantage of using phage vectors is that foreign DNA can be packed into the phage (invitro packaging), the latter in turn can be injected into the host cell very effectively (Note: no transformation is required). Figure 12.28 shows how a λ phage is used for cloning.

Cosmids: Cosmids are the vectors possessing the characteristics of both plasmid and bacteriophage. Cosmids can be constructed by adding a fragment of phage λ DNA including cos site, to plasmids. Once inside the host cell, cosmids behave just like plasmids and replicate. The advantage with cosmids is that they carry larger fragments of foreign DNA (35–45 kb) compared to plasmids.

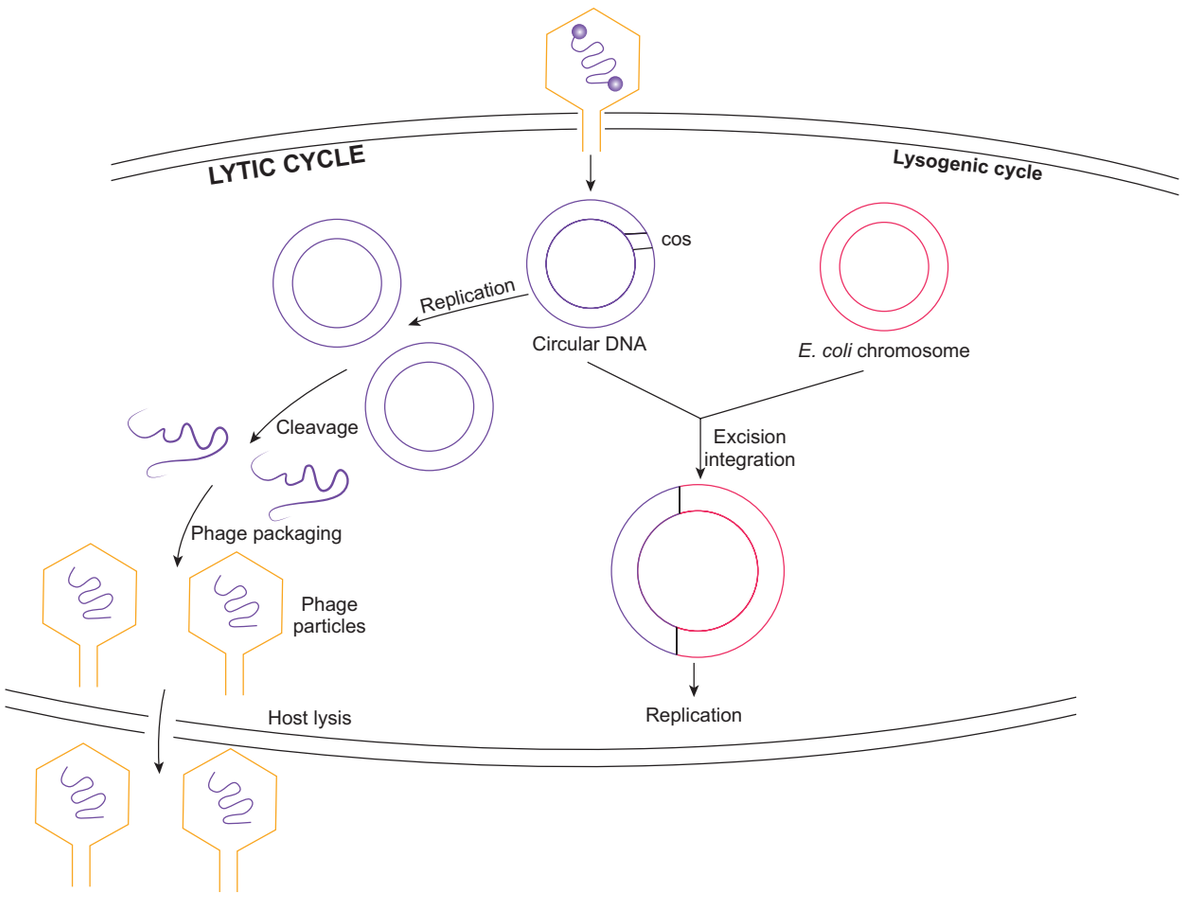


Figure 12.27: Life cycle of phage λ

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